Tasca Avaluable 7.4

1. Carrega la informació dels arxius Transactions.json i Accounts.json al mysql workbench en dues taules. Explica tot el procés amb captures de pantalla i copia el codi.

Pas 1: Crear la base de dades i les taules

CREATE DATABASE banc;

USE banc;

CREATE TABLE transactions (id INT AUTO_INCREMENT PRIMARY KEY, json_data JSON);

CREATE TABLE accounts (id INT AUTO_INCREMENT PRIMARY KEY, json_data JSON);

```
CREATE DATABASE banc;

USE banc;

CREATE TABLE transactions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    json_data JSON
);

CREATE TABLE accounts (
    id INT AUTO_INCREMENT PRIMARY KEY,
    json_data JSON
);
```

Pas 2: carregar-hi les dades JSON

INSERT INTO transactions (json_data) VALUES ('{"txn_id": "1", "amount": 100, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 1, "teller_emp_id": 1, "execution_branch_id": 1, "execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.", "execution_branch_city": "Waltham", "execution_branch_state": "MA", "execution_branch_zip_code": "02451"}'),

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```
INSERT INTO transactions (json_data) VALUES
('{"txn_id": "1", "amount": 100, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 1, "teller_emp_id": 1, "execu
('{"txn_id": "2", "amount": 500, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 2, "teller_emp_id": 2, "execu
('{"txn_id": "3", "amount": 75, "funds_avail_date": "2004-06-30 00:00:00", "txn_date": "2004-06-30 00:00:00", "txn_type_cd": "CDT", "account_id": 3, "teller_emp_id": 2, "execut
('{"txn_id": "4", "amount": 1125, "funds_avail_date": "2001-03-12 00:00:00", "txn_date": "2001-03-12 00:00:00", "txn_type_cd": "CDT", "account_id": 4, "teller_emp_id": 2, "exec
('{"txn_id": "5", "amount": 250, "funds_avail_date": "2001-03-12 00:00:00", "txn_date": "2001-03-12 00:00:00", "txn_type_cd": "CDT", "account_id": 5, "teller_emp_id": 7, "execu
('{"txn_id": "6", "amount": 325, "funds_avail_date": "2002-11-23 00:00:00", "txn_date": "2002-11-23 00:00:00", "txn_type_cd": "CDT", "account_id": 7, "teller_emp_id": 9, "execu
('{"txn_id": "7", "amount": 350, "funds_avail_date": "2002-12-15 00:00:00", "txn_date": "2002-12-15 00:00:00", "txn_type_cd": "CDT", "account_id": 8, "teller_emp_id": 7, "execu
('{"txn_id": "8", "amount": 154, "funds_avail_date": "2003-09-12 00:00:00", "txn_date": "2003-09-12 00:00:00", "txn_type_cd": "CDT", "account_id": 10, "teller_emp_id": 12, "exe
('{"txn_id": "9", "amount": 55, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 11, "teller_emp_id": 9, "execu
('{"txn_id": "10", "amount": 452, "funds_avail_date": "2004-09-30 00:00:00", "txn_date": "2004-09-30 00:00:00", "txn_type_cd": "CDT", "account_id": 12, "teller_emp_id": 13, "ex
```

INSERT INTO accounts (json_data) VALUES ('{"account_id": "1", "avail_balance": 1057.75, "close_date": "NULL", "last_activity_date": "2005-01-04", "open_date": "2000-01-15", "pending_balance": 1057.75, "status": "ACTIVE", "cust_id": 1, "open_emp_id": 10, "open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address": "422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA", "open_branch_zip_code": "01801", "product_cd": "CHK", "product_date_offered": "2000-01-01", "product_date_retired": "NULL", "product_name": "checking account", "product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}')

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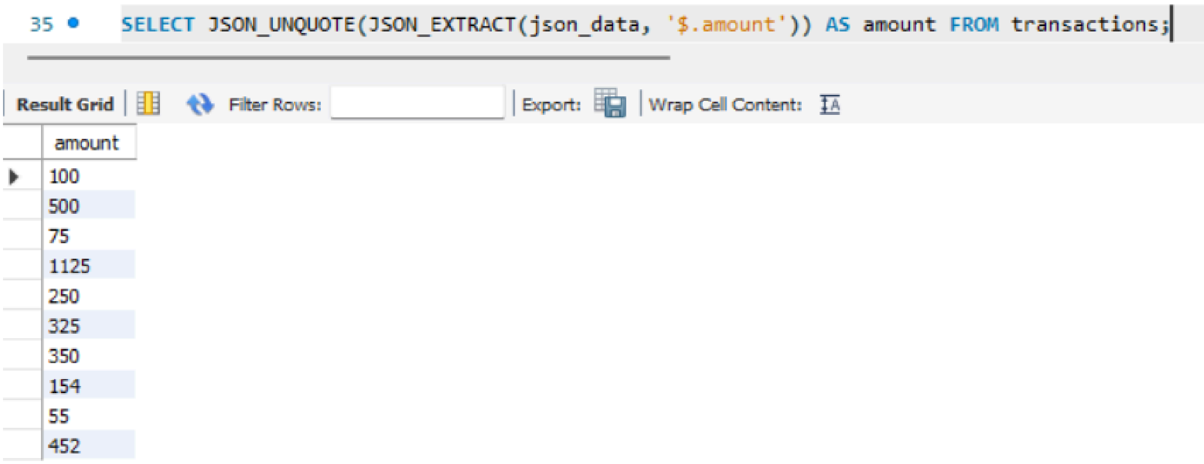
```
INSERT INTO accounts (json_data) VALUES
('{"account_id": "1", "avail_balance": 1057.75, "close_date": "NULL", "last_activity_date": "2005-01-04", "open_date": "2000-01-15", "pending_balance": 1057.75, "status": "ACTI
('{"account_id": "2", "avail_balance": 500, "close_date": "NULL", "last_activity_date": "2004-12-19", "open_date": "2000-01-15", "pending_balance": 500, "status": "ACTIVE", "cu
('{"account_id": "3", "avail_balance": 3000, "close_date": "NULL", "last_activity_date": "2004-06-30", "open_date": "2004-06-30", "pending_balance": 3000, "status": "ACTIVE", "
('{"account_id": "4", "avail_balance": 2258.02, "close_date": "NULL", "last_activity_date": "2004-12-27", "open_date": "2001-03-12", "pending_balance": 2258.02, "status": "ACTI
('{"account_id": "5", "avail_balance": 200, "close_date": "NULL", "last_activity_date": "2004-12-11", "open_date": "2001-03-12", "pending_balance": 200, "status": "ACTIVE", "cu
('{"account_id": "7", "avail_balance": 1057.75, "close_date": "NULL", "last_activity_date": "2004-11-30", "open_date": "2002-11-23", "pending_balance": 1057.75, "status": "ACTI
```

2. Troba la consulta SQL utilitzant les funcions proporcionades per MySQL per a manejar codi JSON que retornaria la solució a les següents preguntes.

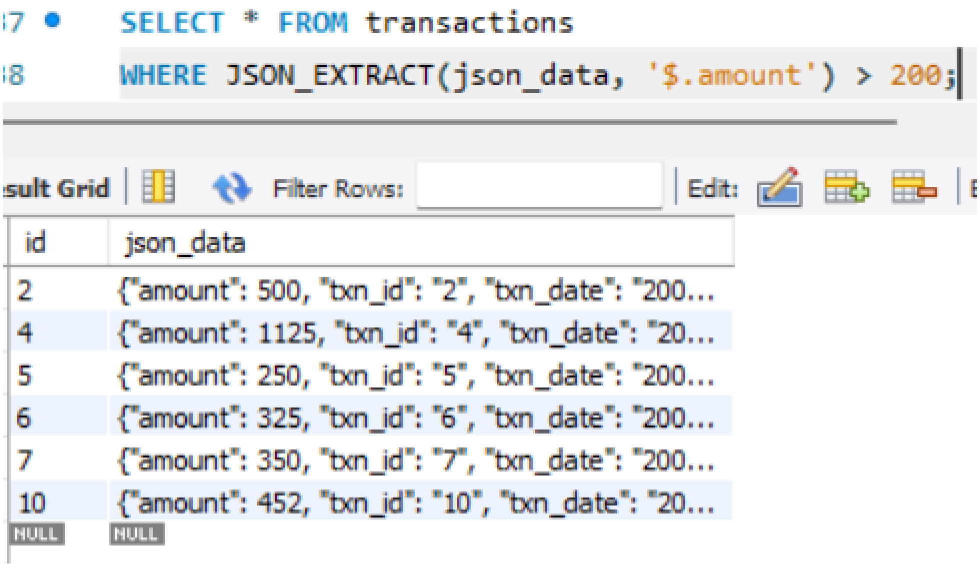
1. Quina consulta utilitzaries per obtenir l'import (amount) de totes les transaccions de la taula transactions (Utilitza la funció JSON_UNQUOTE per a eliminar les cometes)?

SELECT JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.amount')) AS amount FROM transactions;



2. Com podries filtrar les transaccions que tenen una quantitat superior a 200 (utilitza JSON_EXTRACT?

SELECT * FROM transactions
WHERE JSON_EXTRACT(json_data, '$.amount') > 200;



3, Escriu una consulta que obtingui l'ID de la transacció (txn_id) i la data de la transacció (txn_date) per totes les transaccions.

SELECT
JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.txn_id')) AS txn_id,
JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.txn_date')) AS txn_date
FROM transactions;

```
SELECT
JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.txn_id')) AS txn_id,
JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.txn_date')) AS txn_date
FROM transactions;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content:

txn_id	txn_date
1	2000-01-15 00:00:00
2	2000-01-15 00:00:00
3	2004-06-30 00:00:00
4	2001-03-12 00:00:00
5	2001-03-12 00:00:00
6	2002-11-23 00:00:00
7	2002-12-15 00:00:00
8	2003-09-12 00:00:00
9	2000-01-15 00:00:00
10	2004-09-30 00:00:00

4. Escriu una consulta que recuperi l'ID del compte (account_id) i l'import (amount) de les transaccions, unint les taules transactions i accounts per account_id

SELECT

JSON_UNQUOTE(JSON_EXTRACT(a.json_data, '$.account_id')) AS account_id,

JSON_EXTRACT(t.json_data, '$.amount') AS amount

FROM transactions t

JOIN accounts a

ON JSON_UNQUOTE(JSON_EXTRACT(t.json_data, '$.account_id')) =

JSON_UNQUOTE(JSON_EXTRACT(a.json_data, '$.account_id'));

```
SELECT
    JSON_UNQUOTE(JSON_EXTRACT(a.json_data, '$.account_id')) AS account_id,
    JSON_EXTRACT(t.json_data, '$.amount') AS amount
FROM transactions t
JOIN accounts a
  ON JSON_UNQUOTE(JSON_EXTRACT(t.json_data, '$.account_id')) = JSON_UNQUOTE(JSON_EXTRACT(a.json_data, '$.account_id'));
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content:

account_id	amount
1	100
2	500
3	75
4	1125
5	250
7	325

5. Com podries calcular l'import total de les transaccions per cada compte (account_id)?

SELECT JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.account_id')) AS account_id, SUM(CAST(JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.amount')) AS DECIMAL(10,2))) AS total_amount FROM transaction GROUP BY JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.account_id'));

```
SELECT JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.account_id')) AS account_id,
SUM(CAST(JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.amount')) AS DECIMAL(10,2))) AS total_amount
FROM transactions GROUP BY JSON_UNQUOTE(JSON_EXTRACT(json_data, '$.account_id'));
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

account_id	total_amount
1	100.00
2	500.00
3	75.00
4	1125.00
5	250.00
7	325.00
8	350.00
10	154.00
11	55.00
12	452.00